

Evolution from 1G to 5G Wireless Technology





INTRODUCTION

WHAT IS WIRELESS ?

***The word wireless is dictionary defined “having no wires ” .
In networking terminology , wireless is the term used to describe any computer network where there is no physical wired connection between sender and receiver, but rather the network is connected by radio waves and or microwaves to maintain communications.***

Wireless networking utilizes specific equipment such as NICs and Routers in place of wires (copper or optical fibre).

- *1G refers to the first generation of wireless telephone technology, mobile telecommunications which was first introduced in 1980s and completed in early 1990s.*
- *It's Speed was up to 2.4kbps.*
- *It allows the voice calls in 1 country.*
- *1G network use Analog Signal.*
- *Advanced Mobile Phone System (AMPS) was first launched in USA in 1G mobile*
- *AMPS system was frequency modulated and used frequency division multiple access (FDMA) with a channel capacity of 30 KHz and frequency band of 824- 894MHz*



DRAWBACKS OF 1G

- *Poor Voice Quality*
- *Poor Battery Life*
- *Large Phone Size*
- *No Security*
- *Limited Capacity*
- *No Handoff Reliability*
- *Poor security*
- *Offered very low level of spectrum efficiency*
- *FDMA for multiple access*
- *Ex: AMPS*



1G Wireless System



2G TECHNOLOGY

- ❖ *2G technology refers to the 2nd generation which is based on GSM.*
- ❖ *It was launched in Finland in the year 1991.*
- ❖ *2G network use digital signals.*
- ❖ *It's data speed was upto 64kbps.*

Features Includes:

- ✓ *It enables services such as text messages, picture messages and MMS (multi media message).*
- ✓ *It provides better quality and capacity .*
- ✓ *Combination of FDMA and TDMA is used for multiple access*





DRAWBACKS OF 2G

❑ *2G requires strong digital signals to help mobile phones work. If there is no network coverage in any specific area, digital signals would be weak.*

❑ *These systems are unable to handle complex data such as Videos.*

Ex: GSM, DAMPS



2G Wireless System



WIRELESS MODELS OF 1G & 2G

1G WIRELESS SYSTEMS



2G WIRELESS SYSTEMS





2.5G TECHNOLOGY

- ❖ *2.5G is a technology between the second (2G) and third (3G) generation of mobile telephony.*
- ❖ *2.5G is sometimes described as 2G Cellular Technology combined with GPRS.*

Features Includes:

- ✓ *Phone Calls*
- ✓ *Send/Receive E-mail Messages*
- ✓ *Web Browsing*
- ✓ *Speed : 64-144 kbps*
- ✓ *Camera Phones*
- ✓ *Take a time of 6-9 mins. to download a 3 mins. Mp3 song*
- ✓ *Ex: GPRS*





3G TECHNOLOGY

- *3G technology refer to third generation which was introduced in year 2000s.*
- *Data Transmission speed increased from 144kbps- 2Mbps.*
- *Typically called Smart Phones and features increased its bandwidth and data transfer rates to accommodate web-based applications and audio and video files.*





FEATURES OF 3G TECHNOLOGY

- ✓ *Providing Faster Communication*
- ✓ *Send/Receive Large Email Messages*
- ✓ *High Speed Web / More Security*
- ✓ *Video Conferencing / 3D Gaming*
- ✓ *TV Streaming/ Mobile TV/ Phone Calls*
- ✓ *Large Capacities and Broadband Capabilities*
- ✓ *11 sec – 1.5 min. time to download a 3 min Mp3 song.*
- ✓ *WCDMA is used for multiple access*
- ✓ *Ex: UMTS, CDMA 2000*





DRAWBACKS OF 3G TECHNOLOGY

- ◆ *Expensive fees for 3G Licenses Services*
- ◆ *It was challenge to build the infrastructure for 3G*
- ◆ *High Bandwidth Requirement*
- ◆ *Expensive 3G Phones.*
- ◆ *Large Cell Phones*





4G TECHNOLOGY (*Anytime ,Anywhere*)





4G TECHNOLOGY (Anytime ,Anywhere)

- *4G technology refer to or short name of fourth Generation which was started from late 2000s.*
- *Capable of providing 100Mbps – 1Gbps speed.*
- *One of the basic term used to describe 4G is MAGIC.*

MAGIC:

- *Mobile Multimedia*
- *Anytime Anywhere*
- *Global Mobility Support*
- *Integrated Wireless Solution*
- *Customized Personal Services*

Also known as Mobile Broadband Everywhere.



4G (*Anytime, Anywhere*)

- *The next generations of wireless technology that promises higher data rates and expanded multimedia services.*
- *Capable to provide speed 100Mbps-1Gbps.*
- *High QOS and High Security*
- *Provide any kind of service at any time as per user requirements, anywhere.*
- *OFDMA is used*

Features Include:

- *More Security*
- *High Speed*
- *High Capacity*
- *Low Cost Per-bit etc.*



- *Battery uses is more*
- *Hard to implement*
- *Need complicated hardware*
- *Expensive equipment required to implement next generation network*
- *Ex: LTE*





COMPARISON BETWEEN 3G Vs 4G

The basic difference between 3G and 4G is in data transfer and signal quality.

<i>Technology</i>	<i>3G</i>	<i>4G</i>
<i>Data Transfer Rate</i>	<i>3.1 MB/sec</i>	<i>100 MB/sec</i>
<i>Internet Services</i>	<i>Broadband</i>	<i>Ultra Broadband</i>
<i>Mobile - TV Resolution</i>	<i>Low</i>	<i>High</i>
<i>Bandwidth</i>	<i>5-20 MHz</i>	<i>100MHz</i>
<i>Frequency</i>	<i>1.6-2 GHz</i>	<i>2-8 GHz</i>
<i>Download and upload</i>	<i>5.8 Mbps</i>	<i>14 Mbps</i>



WIRELESS MODELS OF 3G & 4G

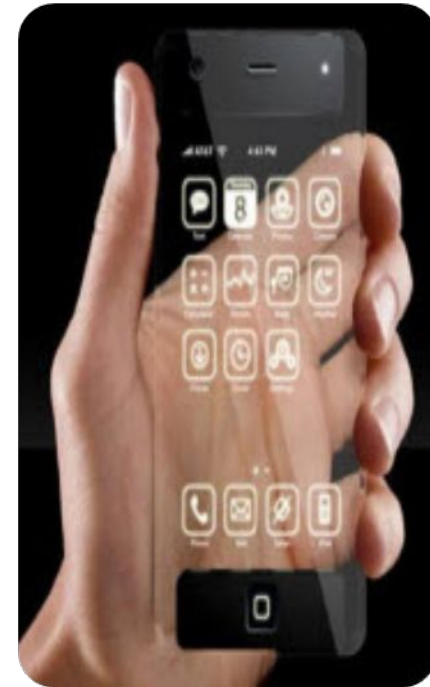
□ 3G WIRELESS SYSTEM



□ 4G WIRELESS SYSTEM



- *5G technology refer to short name of fifth Generation which was started from late 2010s.*
- *Complete wireless communication with almost no limitations.*
- *It is highly supportable to WWW (Wireless World Wide Web).*



- *High Speed, High Capacity*
- *5G technology providing large broadcasting of data in Gbps .*
- *Multi - Media Newspapers, watch T.V programs with the clarity as to that of an HD Quality.*
- *Faster data transmission that of the previous generations.*
- *Large Phone Memory, Dialing Speed, clarity in Audio/Video.*
- *Support interactive multimedia , voice, streaming video, Internet and other*
- *5G is More Effective and More Attractive.*



The following basic differences between 4G and 5G are:

<i>Technology</i>	<i>4G(2000-10)</i>	<i>5G(2010-20)</i>
<i>Switching</i>	<i>Circuit/Packet</i>	<i>Circuit/Packet</i>
<i>Data Rate</i>	<i>Upto 20Mbps</i>	<i>Upto 1 Gbps</i>
<i>Technology</i>	<i>Combination of broadband LAN/WAN/PAN</i>	<i>Combination of broadband LAN/WAN/PAN</i>



EVOLUTION OF 1G TO 5G TECHNOLOGY



1G
1981



2G
1992



3G
2001



4G
2011



5G
2020